

L48 ML and the Physical World

Project Submission

Carl Henrik Ek, Neil D. Lawrence*

November 19, 2024

Abstract

This document outlines the group project submission for L48 Machine Learning and the Physical World.

1 Assessment

1.1 Report

You will submit a report on the moodle page of the course. The format of this should be as a paper. Use the style files and instructions here <https://nips.cc/Conferences/2023/PaperInformation/StyleFiles>. This means that the final submission should be 9 pages of content which should be plenty for you to describe the work and conclusions. Remember, you do not need to explain for example Gaussian process basics if this is something that you use for your project instead spend time explain what you have done and the specifics of your project and if any novelties (say for example interesting kernels) that you have used. Try to focus on the narrative, motivate why you are making specific choices to address the problem at hand.

1.2 Oral

As the final part of the assessment you will have a short oral examination. This presentation and viva is on your own material so do not think of it as a test, it is for you to make sure that we have understood what you have done so that all is not on the report. You will prepare a 5-10 minute introduction to what you have done and then we will discuss so that we make sure we gotten everything. As soon as we have gone through the reports we will send out a link where you can book a presentation/viva time. The viva will take less than 30 minutes in total and it will also be a place for you to get feedback on the work and give us feedback on the course to tell us what we did well and what we can improve upon.

2 Individual

While this is a group project we are also required by the department to collect information on what each individual have done for the project. To do so each group member should submit a up to one page document with the following content.

- the CSRid of all members in your group
- summary of your personal contribution to the work of the team
- your assessment of the contributions made by each other member of the team

*che29@cam.ac.uk, ndl21@cam.ac.uk

3 Final Submission

One member of the team submits the final report and each member of the team submits the document outlining the contributions. Just to reduce my detective work it would be great if you put the CSRid of all members of the group in the report.



Machine Learning and the Physical World

Projects

Carl Henrik Ek - che29@cam.ac.uk

21st of November, 2024

<http://carlhenrik.com>



The End

Week 1 Introduction: ML and the Physical World

Week 1 Quantification of Beliefs

Week 2 Gaussian processes

Week 2 Simulation

Week 3 Emulation

Week 3 Sequential Decision Making Under Uncertainty: Bayesian Optimisation

Week 4 Probabilistic Numerics

Week 4 Emukit and Experimental Design

Week 5 Sensitivity Analysis

Week 5 Multifidelity Modelling





Thursday 21/11 Jack Naish

Tuesday 23/11 Mohammad-Hadi Sotoudeh

Thursday 28/11 Alistair Senior

Tuesday 3/12 Cancelled



Practical

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- Concluded with a report
- Assessed with a **viva** on the report **and** how it relates to the material in the course
- The viva will also be where you get feedback, and can give feedback¹

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- several "loops" of science, you learn one thing and then do a conditional experiments

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 - *look we have achieved SOTA on this*

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- Mail groups to me by the end of next week

Q

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 - get feedback

eof